

FOSSIL FUEL



What are Fossil fuels?

- ▶ Fossil fuels are nonrenewable hydrocarbon containing material formed by natural resources such as anaerobic decomposition of buried dead organisms.
- ▶ Fossil fuels are of great importance because they can be burned and producing significant amounts of energy.

Types of fossil fuel

❖ COAL

❖ OIL

❖ NATURAL GAS



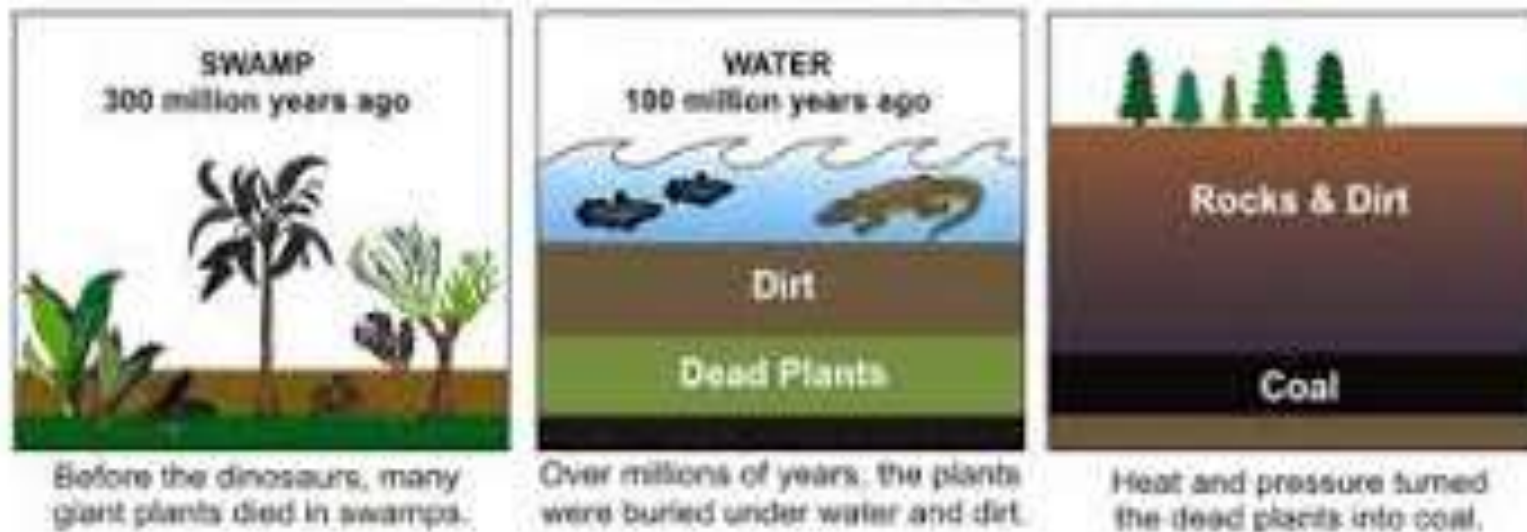


Coal Facts

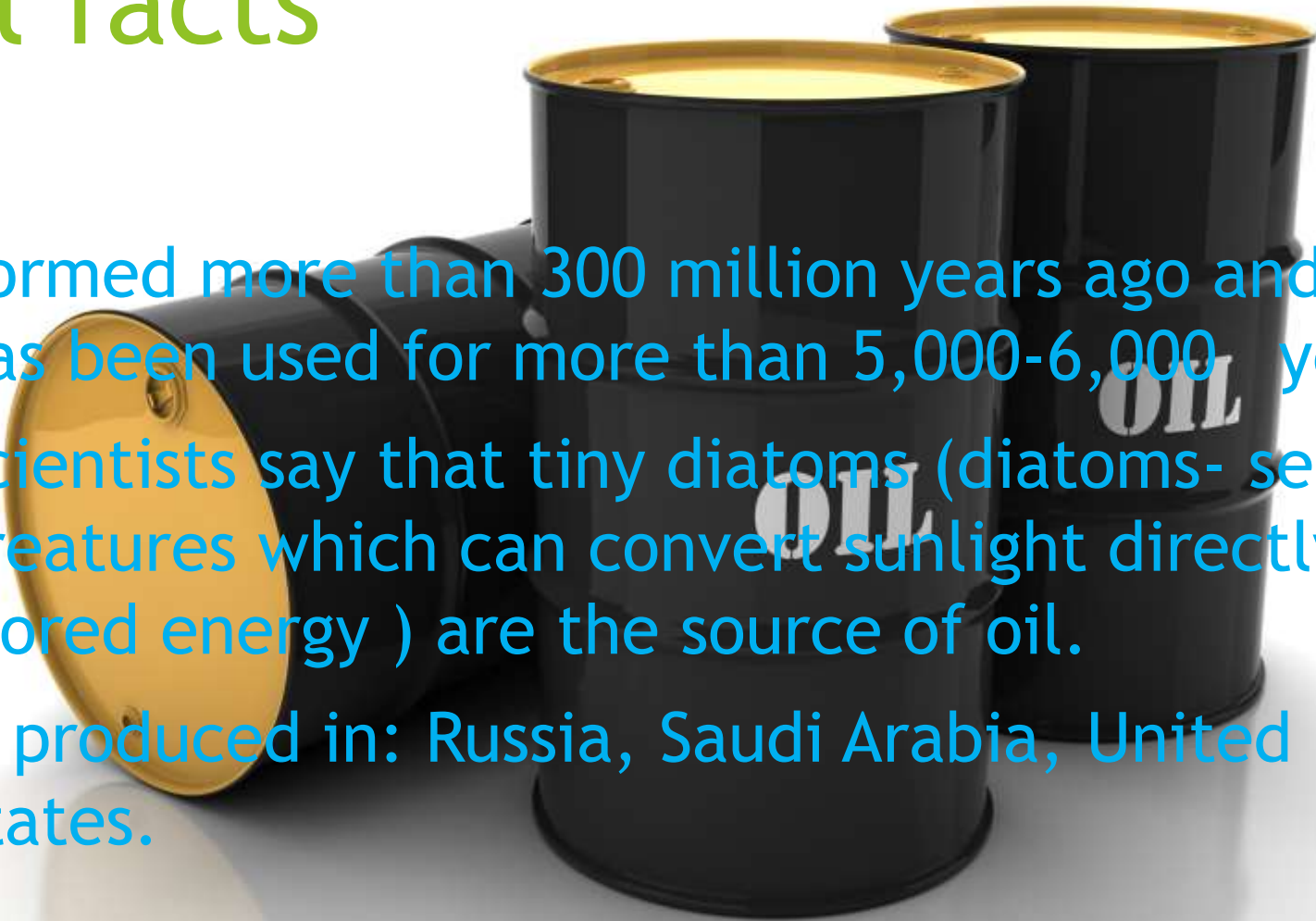
- ✓ Most abundant fossil fuel produced in the United States.
- ✓ A nonrenewable energy source because it takes millions of years to create.
- ✓ Energy in coal comes from the energy stored by plants that lived hundreds of millions of years ago.

Formation of coal

- ▶ Formed when peat is altered physically and chemically.
- ▶ Formed from the remains of vegetation that grew as long as 400 million years ago.
- ▶ Process is called "coalification."



Oil facts

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- ▶ Formed more than 300 million years ago and it has been used for more than 5,000-6,000 years.
 - ▶ Scientists say that tiny diatoms (diatoms- sea creatures which can convert sunlight directly into stored energy) are the source of oil.
 - ▶ Is produced in: Russia, Saudi Arabia, United States.

Formation of oil

- ▶ Oil was formed from the remains of animals and plants(diatoms) that lived millions of years ago in a marine (water) environment before the dinosaurs.

Natural gas facts

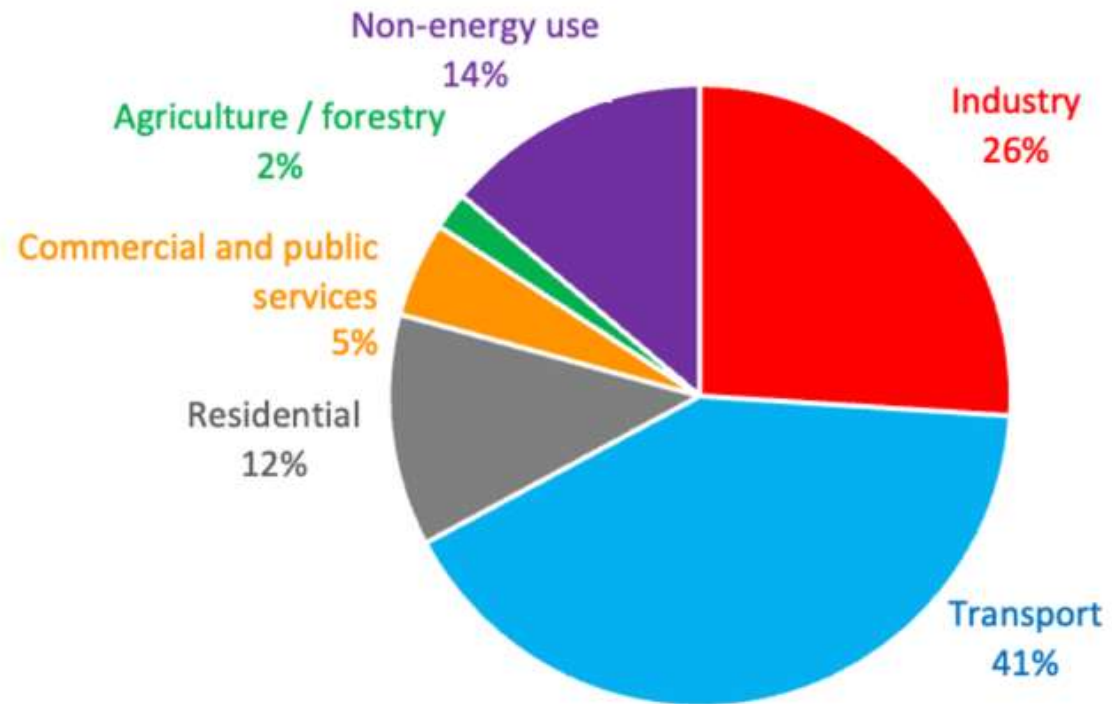
- ▶ Lighter than air , no odor.
- ▶ First discoveries of natural gas seeps were made in Iran.
- ▶ Mostly made up of a gas called methane.
- ▶ Usually found near petroleum underground and pumped from below ground and travels in pipelines to storage areas.

Formation of natural gas

- ▶ Methane, a gas or compound that has one carbon atom and four hydrogen atoms.

Fossil Fuel Consumption sector:

World Fossil Fuel Consumption by Sector, 2019

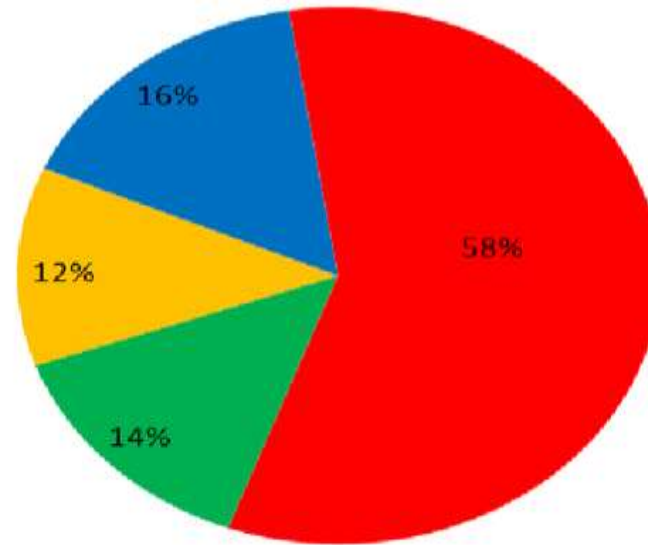


Uses of Coal

- ▶ Industries.
- ▶ Domestic Use.
- ▶ Gasification and Liquefaction.
- ▶ Almost 93% of the coal used in the United States is used for generating electricity.

USES OF COAL

Coal usage



■ Residential, agricultural, etc.

■ Cement and other industrial applications

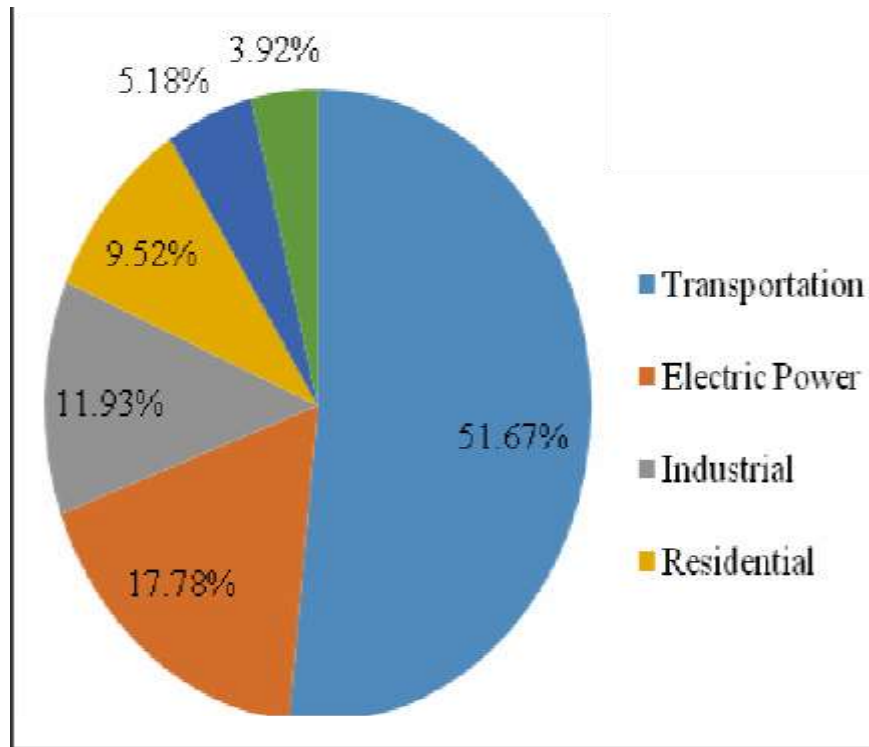
■ Metallurgy (Iron and steel making)

■ Electricity and heat generation

Uses of Oil

- ▶ To propel vehicles
- ▶ To produce electricity
- ▶ In industrial areas
- ▶ For residential use

USES OF OIL



Uses Of Natural Gas

- ▶ Raw material to produce chemicals, fertilizer and hydrogen.
- ▶ To heat building and water , to dry clothes.
- ▶ To operate refrigeration and cooling equipment.
- ▶ For transportation services.

Disadvantages of fossil fuel

- ▶ Contribute to climate change.
- ▶ Non-renewable.
- ▶ Unsustainable.
- ▶ Hard to extract.
- ▶ Accident-prone.



Exxon valdez oil

- ▶ Unprecedented underwater leak spreads oil across the northern gulf of Mexico between the mouth of the Mississippi river and Florida.
- ▶ Coast Guard says remote underwater cameras detect the well is leaking 1,000 barrels of crude oil per day.
- ▶ The Coast Guard says the leak will affect the Gulf shore.
- ▶ U.S. official close areas affected by the spill to fishing for an initial period of 10 days.

Deep water horizon

- ▶ Deep Water Horizon oil rig exploded in the Gulf of Mexico
- ▶ There is some progress cleaning it up
- ▶ Last update showed that it was getting closer to the coast of Louisiana.
- ▶ Reach more than the half of the size of the valdez disaster.
- ▶ more than 1 million gallons of water has been cleaned from oil and 367881 feet of oil booms have been deployed.
- ▶ BP has been trying all sorts of engineering efforts to stop the leak, which includes the four story high containment dome which was to be placed over the leakage. |

ALTERNATIVE OF FOSSIL FUEL

- ▶ Renewable Energy.
- ▶ Nuclear Power.
- ▶ Hydrogen.
- ▶ Biomass.
- ▶ Geothermal energy.



References

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